

Project Design Phase

Solution Architecture

Date	22 September 2026
Team ID	SWTID-2026-8694
Project Name	AI BLOGNEST API
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. For AI BLOGNEST API, the architecture describes the structure, characteristics, behavior and key technology components of the backend solution.

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Architecture and Data Flow of AI BLOGNEST API

CLIENT / API TESTING Web or application client • Postman / Thunder Client
API LAYER HTTP Request → Express.js Routes → Controllers
BUSINESS / VALIDATION LAYER Request validation • Authentication-related checks • Error handling
DATA LAYER Mongoose Models / Schemas → MongoDB Database
RESPONSE FLOW MongoDB result → Controller → JSON Response → Client

Technology Components

Layer	Technology	Role
Runtime	Node.js	Runs the backend application
API Framework	Express.js	Routes requests and sends responses
Database	MongoDB	Stores application/blog data
ODM	Mongoose	Defines schemas and communicates with MongoDB
Testing	Postman / Thunder Client	Tests API endpoints and responses

Figure 1: Architecture and data flow of the AI BLOGNEST API

Reference:

<https://aws.amazon.com/blogs/industries/voice-applications-in-clinical-research-powered-by-ai-on-aws-part-1-architecture-and-design-considerations/>